

KUSHIRO KENEBETSU, Japan

WMO: 474180

Lat: 42.9861N

Lon: 144.3772E

Elev: 131

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.8	8.4	-2.8	4.8	10.0	0.3	5.6	12.9	36.3	29.5	32.1	30.9	8.0	20	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.1	75.0	68.8	72.6	67.7	70.7	66.7	70.7	73.2	69.1	71.6	67.6	70.0	10.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.9	110.5	71.9	68.1	103.9	70.6	66.6	98.5	69.4	34.8	73.4	33.4	71.9	32.2	70.0	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
32.3	27.6	23.8	-1.0	81.4	4.2	3.2	-4.1	83.7	-6.5	85.6	-8.9	87.4	-12.0	89.7		
			-1.7	72.6	4.1	1.9	-4.7	73.9	-7.1	75.0	-9.4	76.1	-12.4	77.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.9	22.6	23.1	30.9	39.4	47.7	54.4	61.7	65.4	61.6	51.1	39.7	28.1
	DBStd	15.67	5.77	6.15	5.44	4.67	5.03	4.75	4.49	4.42	4.89	5.50	6.40	6.15
	HDD50	3679	849	754	593	318	105	13	0	0	0	54	313	680
	HDD65	7802	1314	1174	1058	768	537	321	121	50	125	430	759	1145
	CDD50	1458	0	0	0	1	33	144	363	478	347	89	3	0
	CDD65	106	0	0	0	0	0	1	19	63	23	0	0	0
	CDH74	124	0	0	0	0	0	4	19	79	22	0	0	0
	CDH80	7	0	0	0	0	0	1	0	5	1	0	0	0
	WSAvg	10.4	11.0	11.3	11.8	10.9	10.3	8.5	7.7	8.3	9.3	11.2	12.3	11.8
Precipitation	PrecAvg	43.7	1.7	1.1	2.0	3.3	4.7	4.3	4.9	5.5	6.5	4.2	2.8	2.5
	PrecMax	61.8	5.2	3.1	4.4	8.1	10.0	11.3	11.2	17.0	11.0	9.7	6.8	5.8
	PrecMin	32.4	0.2	0.2	0.3	0.5	0.9	1.1	0.8	1.3	2.6	1.0	0.3	0.5
	PrecStd	7.5	1.2	0.9	1.1	2.0	2.2	2.8	2.5	3.3	2.3	2.3	1.6	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.1	39.9	47.3	57.1	66.9	71.1	76.5	80.0	76.9	67.4	57.7	48.9
		MCWB	37.3	35.3	39.7	45.5	54.5	62.8	68.5	71.2	70.1	59.2	52.9	45.9
	2%	DB	36.9	37.3	42.9	52.3	61.3	66.6	73.1	75.9	72.9	64.4	56.0	45.2
		MCWB	34.0	33.8	37.5	44.0	52.0	60.1	66.8	69.7	67.5	58.4	51.9	41.8
	5%	DB	34.9	35.0	40.4	49.1	57.8	63.2	70.4	73.4	70.9	62.5	53.9	41.7
		MCWB	31.6	31.8	35.9	42.8	50.5	58.3	65.8	68.6	66.6	57.5	49.9	37.4
	10%	DB	33.1	33.3	38.5	46.6	55.0	60.9	68.2	71.5	69.1	60.5	51.8	38.6
		MCWB	29.6	30.3	34.6	41.9	49.8	57.1	65.0	68.2	64.9	55.8	47.7	34.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.1	37.2	41.9	48.6	57.1	64.5	70.1	72.9	72.5	63.1	55.5	46.7
		MCDB	39.3	38.5	45.3	52.9	62.7	69.0	73.5	76.1	74.3	65.0	56.8	47.9
	2%	WB	34.9	34.2	39.0	45.9	54.2	61.4	68.2	71.2	70.0	60.6	53.2	42.4
		MCDB	36.5	36.1	41.8	49.8	58.4	65.0	71.0	73.8	71.9	62.8	55.1	44.7
	5%	WB	32.3	32.5	36.9	44.2	52.2	59.3	66.8	70.0	67.8	58.8	50.7	38.2
		MCDB	34.3	34.7	39.4	47.6	55.8	61.9	69.5	72.5	69.8	61.2	53.2	41.0
	10%	WB	29.7	30.8	35.1	42.6	50.6	57.7	65.3	68.6	65.7	56.9	48.0	34.7
		MCDB	32.8	33.3	37.8	46.0	54.0	60.0	67.8	71.0	68.2	59.7	51.1	37.9
Mean Daily Temperature Range	5% DB	MDBR	16.3	15.8	12.2	11.7	11.6	9.9	9.2	9.1	11.3	14.7	15.0	15.4
		MCDBR	16.0	14.2	12.8	15.7	16.9	13.8	13.0	11.7	12.2	14.2	16.5	17.4
		MCWBR	14.4	12.1	9.7	9.7	9.4	8.4	8.0	6.8	7.7	10.8	14.9	16.0
	5% WB	MCDBR	16.3	13.9	12.3	14.0	13.9	11.9	11.4	10.2	9.8	12.5	16.0	17.7
		MCWBR	16.1	12.9	10.3	10.0	9.0	8.1	7.7	6.6	7.1	11.0	15.9	17.3
Clear-Sky Solar Irradiance	taub		0.322	0.368	0.446	0.505	0.522	0.496	0.483	0.466	0.412	0.394	0.368	0.334
	taud		2.186	2.037	1.919	1.848	1.891	2.053	2.159	2.224	2.330	2.245	2.224	2.204
	Ebn at Noon		250	254	247	243	243	250	252	252	258	245	230	232
	Edh at Noon		30	42	54	62	61	52	47	42	35	34	30	28
All-Sky Solar Radiation	RadAvg		631	922	1245	1468	1551	1530	1391	1287	1135	919	625	512
	RadStd		43	60	87	109	133	136	126	115	98	59	43	39

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.76	N/A	N/A	N/A	N/A	+1.40	N/A	-211	+155	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-119	N/A	N/A

Nomenclature: See separate page